

2 Logic

2.1 Statements

Decide whether or not the following are statements. In the case of a statement, say if it is true or false, if possible.

- 2. Every even integer is a real number. $\rightarrow$ Statement, true.
- 4. Sets $\mathbb{Z}$ and $\mathbb{N}$. $\rightarrow$ Not a statement.
- 6. Some sets are finite. $\rightarrow$ Statement, true.
- 8. $\mathbb{N} \notin \mathcal{P}(\mathbb{N}) \rightarrow$ Statement, false.
- 10. $(\mathbb{R} \times \mathbb{N}) \cap (\mathbb{N} \times \mathbb{R}) = \mathbb{N} \times \mathbb{N}$. $\rightarrow$ Statement, true.
- 12. If the integer x is a multiple of 7, then it is divisible by 7. $\rightarrow$ Statement, true.
- 14. Call me Ishmael. $\rightarrow$ Not a statement.